

Educational Technologies

Clinical Simulation

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1 Project Summary

The Clinical Simulation is an interactive, web-based eLearning designed to bridge the gap between theoretical medical knowledge and practical clinical reasoning. It places the user in a simulated Emergency Room environment where they must diagnose patients presenting with authentic, ill-structured clinical scenarios.

The platform guides users through a strict four-stage clinical workflow: Anamnesis (patient history), Diagnostics (ordering lab tests and imaging within a budget), Diagnosis (free-text articulation of the clinical hypothesis), and Feedback. By integrating real-time formative feedback, dynamic difficulty scaling (fading budgets), and comprehensive metacognitive analytics, the application is designed to support and foster structured, cost-conscious clinical reasoning among learners.

(Link to the eLearning)

2 Target Audience

The primary target audience consists of medical students, specifically those transitioning from pre-clinical (theoretical) studies to the clinical phase of their education.

Novice Students benefit heavily from the application’s scaffolding (structured workflows and generous starting budgets) and formative feedback, allowing them to make mistakes and learn in a risk-free sandbox.

Advanced Students are challenged by the system’s “fading” mechanics, where consistent success leads to tighter diagnostic budgets. This forces them to refine their diagnostic efficiency and clinical precision to expert levels.

2.1 Learning Outcomes

Upon completing a full 10-trial block, students should be able to gather a targeted patient history and justify which anamnesis questions are diagnostically relevant for a given chief complaint. They should be able to select diagnostic tests in a deliberate, hypothesis-driven manner rather than ordering exhaustively, and to synthesize subjective symptoms and objective findings into a single, articulated clinical diagnosis. Finally, they should be able to reflect on their own diagnostic patterns across multiple cases and identify inefficiencies such as redundant or irrelevant testing.

3 Educational Concepts

3.1 Problem-Based Learning (PBL)

Concept: Learning driven by complex, open-ended problems rather than direct instruction (Barrows, 1986).

Why it is effective here: In real-world medicine, patients present with symptoms (ill-structured problems), not clear labels. By forcing students to synthesize a diagnosis from chief complaints and diagnostic tests, the platform cultivates independent clinical reasoning and mimics genuine medical practice far better than traditional rote memorization or multiple-choice quizzes.

3.2 Cognitive Apprenticeship & Modeling

Concept: Novices acquire cognitive skills in a situated environment under expert guidance, making the expert's thought processes visible (Collins, Brown, & Newman, 1989).

Why it is effective here: The Emergency Room is a high-stress environment where expert decision-making is often internal and invisible. By showing the "Expert's Path" on the feedback screen, revealing exactly which tests an experienced, cost-conscious physician would have ordered, the simulation explicitly models expert efficiency, allowing students to compare and adjust their own cognitive paths.

3.3 Scaffolding (Structural and Boundary)

Concept: Temporary support structures that assist learners in completing tasks they cannot yet do independently (Wood, Bruner, & Ross, 1976), which simultaneously helps manage cognitive load (Sweller, 1988).

Why it is effective here: Novice medical students often experience cognitive overload and resort to "shotgun-testing" (randomly ordering all available tests). The strict 4-stage UI (Structural Scaffolding) enforces a professional workflow (history must be taken before diagnostics). The starting diagnostic budget (Boundary Scaffolding) forces deliberate, hypothesis-driven choices rather than random guessing.

3.4 Adaptive Fading

Concept: The gradual removal of scaffolding as the learner's expertise increases, known as the guidance fading effect (Collins et al., 1989; Renkl & Atkinson, 2003).

Why it is effective here: To ensure students remain in their Zone of Proximal Development (Vygotsky, 1978), the application dynamically reduces the starting budget (decreasing by \$100 per successfully solved case, down to a minimum of \$500). This ensures that the platform remains challenging as clinical competence grows, pushing the student from a forgiving exploratory phase toward strict expert precision.

3.5 Formative Feedback & Active Experimentation

Concept: Developmental, ongoing feedback during the learning process (Black & Wiliam, 1998) combined with safe hypothesis testing (Kolb, 1984).

Why it is effective here: An incorrect diagnosis in reality can harm a patient. In this simulation, submitting a wrong diagnosis does not result in a punitive "Game Over." Instead, it triggers an explanatory feedback card and a "Return to Patient" loop. This iterative process allows students to immediately apply the feedback, order targeted tests to correct their hypothesis, and learn deeply from their mistakes in a zero-risk environment.

3.6 Articulation & Active Recall

Concept: Forcing learners to externalize their implicit reasoning (Collins et al., 1989).

Why it is effective here: The simulation uses a free-text input field for the final diagnosis instead of a multiple-choice list. Multiple-choice formats only test passive recognition (guessing based on seeing the answer). Free-text entry forces active cognitive recall and requires the student to commit to a synthesized clinical hypothesis. While realistically the system evaluates this via keyword string-matching rather than full semantic analysis, the pedagogical value lies in forcing the student to actively retrieve and formulate the diagnosis, simulating the articulation needed in a real medical chart.

3.7 Summative Assessment

Concept: Evaluating learning at the end of a unit against a benchmark (Scriven, 1967).

Why it is effective here: The final score penalizes students for ordering irrelevant tests. This accurately reflects real-world healthcare economics, teaching students that clinical excellence involves not only accuracy but also minimizing patient cost, strain, and wait times.

3.8 Learning Analytics & Metacognitive Reflection

Concept: Using performance data to foster self-regulation and systematically reviewing past actions to improve future performance (Siemens & Long, 2011; Schön, 1983).

Why it is effective here: The “10-Trial Block Evaluation Dashboard” provides a visual ledger of all completed cases, color-coding relevant (green) vs. irrelevant (red) tests. This allows students to conduct a “cognitive audit” of their own performance. By visualizing financial waste and diagnostic search patterns over time, it bridges the gap between active simulation gameplay and improved reflective practice.

4 Technical Implementation

The platform is implemented as a zero-install, client-side HTML5 application that runs in any modern browser, requiring no server-side setup or user accounts. This design choice lowers the barrier to access and supports the goal of an immediately available learning sandbox.

Application state, including the current trial, accumulated score, diagnostic expenditure, and case history, is persisted across page refreshes, ensuring that a 10-trial block can be completed across multiple sessions without data loss. Each block shuffles the available cases to ensure exposure to all ten clinical conditions while varying their order.

The final diagnosis is evaluated through string parsing: the free-text submission is normalized and matched against a set of accepted target answers stored for each case. It is worth noting that this is a keyword-based matching approach rather than true natural-language understanding; the pedagogical value lies in forcing active recall and commitment to a hypothesis, while the evaluation itself remains a deterministic comparison against predefined correct answers.

5 Limitations and Future Work

While the platform integrates a range of established educational concepts, several limitations should be acknowledged.

First, the diagnosis evaluation relies on keyword matching rather than semantic analysis. This means valid answers phrased in unexpected ways may be marked incorrect, and conversely, the system cannot assess the quality of clinical reasoning beyond the presence of target terms.

Second, the case library is currently limited to ten conditions. A broader and more varied case base would be necessary to cover the diagnostic breadth encountered in real clinical practice and to reduce the predictability of repeated blocks.

Third, the application has not yet undergone empirical evaluation with the target audience. The didactic effectiveness is grounded in established learning theory but remains to be validated through user studies measuring actual learning outcomes, diagnostic improvement, and student acceptance.

Future work could address these points by introducing semantic answer evaluation, expanding the case library with input from clinical experts, and conducting a controlled study comparing learning gains against traditional instruction.

References

- Barrows, H. S. (1986). A taxonomy of problem-based learning methods. *Medical Education*, 20(6), 481–486.
- Black, P., & Wiliam, D. (1998). Assessment and classroom learning. *Assessment in Education: Principles, Policy & Practice*, 5(1), 7–74.
- Collins, A., Brown, J. S., & Newman, S. E. (1989). Cognitive apprenticeship: Teaching the crafts of reading, writing, and mathematics. In L. B. Resnick (Ed.), *Knowing, learning, and instruction: Essays in honor of Robert Glaser* (pp. 453–494). Lawrence Erlbaum Associates.
- Kolb, D. A. (1984). *Experiential learning: Experience as the source of learning and development*. Prentice Hall.
- Renkl, A., & Atkinson, R. K. (2003). Structuring the transition from example study to problem solving in cognitive skill acquisition: A cognitive load perspective. *Educational Psychologist*, 38(1), 15–22.
- Schön, D. A. (1983). *The reflective practitioner: How professionals think in action*. Basic Books.
- Scriven, M. (1967). The methodology of evaluation. In R. W. Tyler, R. M. Gagné, & M. Scriven (Eds.), *Perspectives of curriculum evaluation* (pp. 39–83). Rand McNally.
- Siemens, G., & Long, P. (2011). Penetrating the fog: Analytics in learning and education. *EDUCAUSE Review*, 46(5), 30–40.
- Sweller, J. (1988). Cognitive load during problem solving: Effects on learning. *Cognitive Science*, 12(2), 257–285.
- Vygotsky, L. S. (1978). *Mind in society: The development of higher psychological processes*. Harvard University Press.
- Wood, D., Bruner, J. S., & Ross, G. (1976). The role of tutoring in problem solving. *Journal of Child Psychology and Psychiatry*, 17(2), 89–100.